Professional Program in

Data Science, Machine Learning, AI & GenAI

eict IIT Roorkee

E&ICT, IIT Roorkee

The IoT Academy
Connecting The Unconnected

Curriculum

Duration: 10 Months | 240 Hours

DS | ML | DL | AI | Generative AI

Advanced Gen AI, Agentic AI & LLMs Are Covered



About The IIT Roorkee

Indian Institute of Technology Roorkee (IIT Roorkee), one of the oldest technical institutions in Asia, was established in **1847** as the Thomason College of Civil Engineering and was later converted to an IIT in 2001

The institute offers a wide range of academic programmes including BTech, BArch, BS, MTech, MSc, MBA, MDes, and PhD across its 23 academic departments, centres, and schools covering engineering, sciences, architecture, management, humanities, and social sciences.

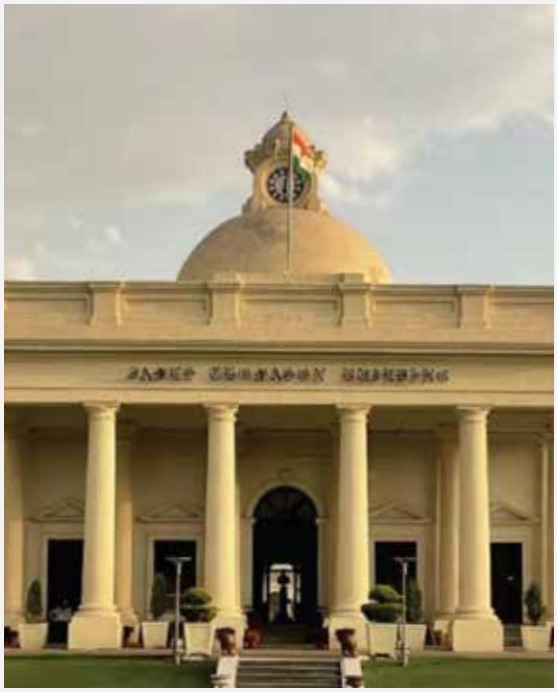
Over the years, **IIT Roorkee** has developed state-of-the-art infrastructure, a strong research ecosystem, and a rich legacy of excellence in technical education and innovation. The institute continues to be recognized nationally and internationally for its academic rigor, research output, and contributions to industry and society.



About The E&ICT Academy IIT Roorkee

Electronics & ICT Academy, IIT Roorkee aims to strengthen the skills of faculty members, working professionals, and students from Engineering, Science, Commerce, and Polytechnic institutions through specialized short-term training programs.
The Academy focuses on both fundamental and advanced areas of Information Technology, Electronics & Communication, Product Design, and Manufacturing.

In addition, E&ICT Academy, IIT Roorkee organizes customized industry-oriented training programs, faculty development programs (FDPs), and research promotion workshops to bridge the gap between academia and industry.
These initiatives are designed to enhance technical competency, encourage innovation, and promote collaborative research in emerging technologies.



About The IoT Academy

The IoT Academy, established in 2017 is a fast emerging company imparting quality programs for skills training, internship and guidance in cutting edge technologies like Data Science, Machine Learning, Artificial Intelligence, Internet of Things, Embedded Systems & many more and focused on helping people develop the skills they need to thrive in the rapidly growing digital economy.

The IoT Academy has collaborated with various premier institutes e.g. E&ICT Academy, IIT-Roorkee, IIT-Guwahati and IIT-Kanpur for Advance Certification courses to take provide outcome-centric solutions to help them achieve their professional goals.

400+
Hiring Partners

30,000+
Learners

8+ Years
Of excellence

81%
Salary Hike

91%
Happy Outcomes





About The Program

Most of the modern technological innovations you see today are powered by Data Science, Machine Learning, AI, and Generative AI. This 10-months Professional Program In Data Science, Machine Learning, AI & GenAI is thoughtfully designed by E&ICT Academy, IIT Roorkee, in collaboration with industry experts, to help you master the exact skills the market demands. The primary goal of this certification program is to prepare you for the most in-demand roles in Data Science, Machine Learning, AI, and Generative AI. You will gain expertise in Artificial Intelligence & Data Technologies, Advanced Analytics, Intelligent Systems, Machine Intelligence, Prompt Engineering and AI/ML-driven Solutions through multiple real-world projects in Machine Learning, Deep Learning, and Generative AI — transforming you into a highly skilled and industry-ready New Age DS, ML, AI & Gen AI Specialist.

Integrated With Gen AI, Agentic AI & LLMs

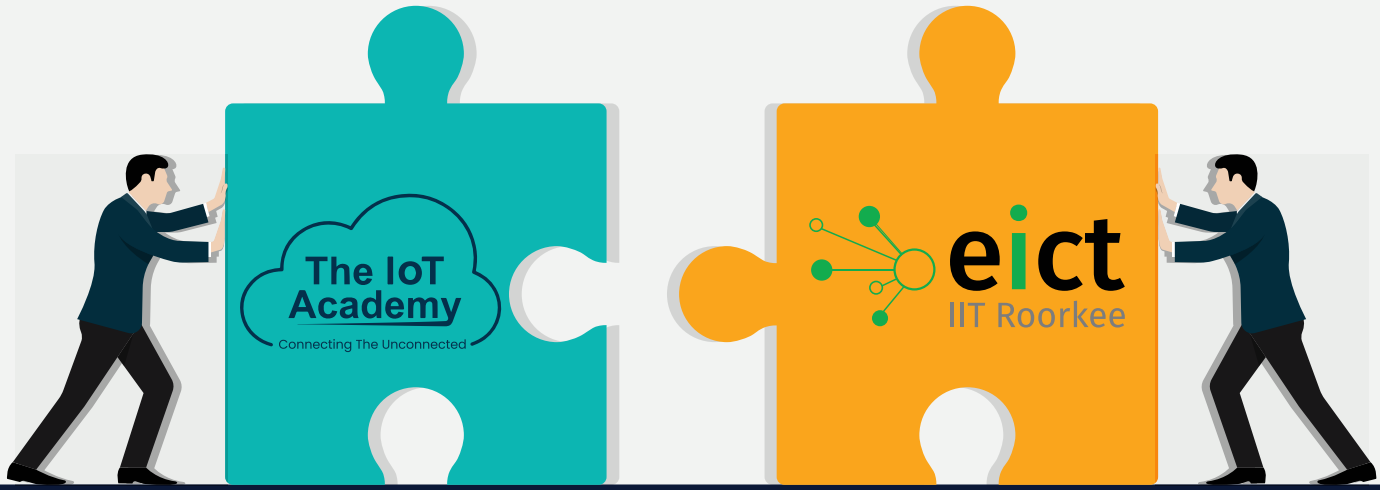
- Gen AI & Agentic AI Modules & Architecture
- Building LLM Applications
- Gen AI & Agentic AI Tools & Case Studies

PARTNERSHIP



About The Partnership

Professional Program In Data Science, Machine Learning, AI & GenAI is offered jointly by The IoT Academy and E&ICT Academy, IIT Roorkee to prepare the future workforce in Machine Learning and AI



Track Curriculum

Module 1 Module 2 Module 3 Module 4 Module 5

Data Analytics, Data Science & Visualization with Python (48 hours 16 Session)

Python Basics

- What is Python
- Application of Python
- Why use Python for AIML Installation
- Anaconda/Other Idle
- Idle to IDE
- Lambda Functions
- Data Types in Python
- Conditional Statement
- Loops in Python
- Functions in Python
- Advance Functions
- File Handling
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Intro to Git & Github and Kaggle

- Git and GitHub
- kaggle

Module 1 Module 2 Module 3 Module 4 Module 5

Python NumPy & Pandas

- Introduction to NumPy
- Exploring a NumPy Array
- Indexing & Slicing a NumPy Array
- Manipulating a NumPy Array
- Performing Mathematical & Statistical Functions using NumPy
- Performing Linear Algebra Operations using NumPy
- Introduction to Pandas
- Exploring Pandas Series
- Introduction to Pandas DataFrame
- Importing & Exporting Data
- Implementing basic DataFrame functionalities
- Exploring Descriptive Statistics with Pandas
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

R Language Essentials

- Fundamentals of R
- Vectors & control statements
- Functions in R
- Matrices & strings
- Lists & arrays in R
- Data visualization in R

Module 1 Module 2 Module 3 Module 4 Module 5

Exploratory Data Analysis (EDA)

- Data Types
- Dispersion & Skewness
- Data imputation
- Data Pre-processing
- Data Cleaning
- Data Manipulation
- Advanced Manipulation
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

SQL For Data Analytics

- SQL Basic
- SQL Joins
- SQL Aggregations
- Subqueries and Temp Tables
- SQL Data Cleaning
- Window Functions
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Data Analysis with Excel

- Key Formulas and Functions, Ranges and Tables
- Data Cleaning – Text Functions, Dates and Times
- Conditional Formatting, Sorting and Filtering
- Dashboard Creation
- Analysis with Pivot Tables
- Data Analysis in Excel – Trends and Patterns
- Data Visualization in Excel – Charts and Plots
- Functions in Python
- Advance Functions File Handling
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Data Visualization with Python

- Why Data Visualization?
- Introduction to Data Visualization
- Libraries & Tools for Data Visualization in Python
- Static Data Visualization Using Seaborn
- Interactive Data Visualization Using Plotly Express
- Interactive Animations & Facet Plots

Module 1 Module 2 Module 3 Module 4 Module 5

Tableau for Business Intelligence

- Understanding Data
- Creating Your First visualization
- Tableau Calculations
- Formatting Visualizations
- Manipulating Data in Tableau
- Creating Dashboards AND Stories
- Distributing & Publishing Your Visualization
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 Module 5

Visual Storytelling using Power BI

- Introduction To Power BI
- Creating Power BI Reports, Auto Filters
- Report Visualization And Properties
- Chart And Map Report Properties
- Hierarchies And Drilldown Reports
- Power BI Deployment & Cloud
- Improving Power BI Reports And More...
- **Mini Project**

Capstone Project- 1 in Data Analytics

Module 1 **Module 2** Module 3 Module 4 Module 5

Applied Maths, Statistics and Probability for ML & AI (21 Hours 7 Sessions)

Mathematics for Machine Learning & AI

- Linear Algebra
- Multi-variable Calculus
- Introduction to Calculus
- Scalar and Vectors
- Text Embeddings
- Gradient Descent

Module 1 **Module 2** Module 3 Module 4 Module 5

Statistics for Machine Learning & AI

- Applications of Statistics
- Introduction to Statistics
- Categories of Data
- Basic Terminologies in Statistics
- Sampling techniques
- Descriptive Statistics
- Measure Used in Descriptive Statistics
- Z-Scores
- Measures of central Tendency
- Feature Scaling
- Skewness

Module 1 **Module 2** Module 3 Module 4 Module 5

Probability

- What is Probability?
- Rules of probability
- Types of Probability
- Random Variables
- Probability Distribution Functions
- Mini Project

Module 1 **Module 2** Module 3 Module 4 Module 5

Inferential Statistics

- Introduction to Inferential Statistics
- Hypothesis Testing
- Normal Distribution
- P-value
- One-tailed and Two-tailed tests
- One Sample Z test
- One Sample T test
- Independent Sample T test
- Chi-square test
- ANOVA

Capstone Project- 2

Module 1 Module 2 **Module 3** Module 4 Module 5

Machine Learning Specialization (60 Hours 20 Sessions)

Introduction to Machine Learning

- What is Machine Learning?
- Applications of Machine Learning
- Machine Learning in your daily life
- Machine Learning in Retail
- Steps Involved in Machine Learning

Module 1 Module 2 **Module 3** Module 4 Module 5

Regression

- Introduction to Regression
- Linear Regression
- Evaluation Metrics in Regression Models
- Logistic Regression
- **Mini Project**

Supervised Classification

- Why Use Classification?
- Application of Classification Algorithms
- Introduction to Classification
- Types of Classification Algorithms
- Classification: Decision Tree
- Classification: Random Forest
- ML in Banking & Finance - Benefits
- Classification: SVM
- Classification: KNN
- Classification: Naïve Bayes
- Evaluating Classification Models
- Model Optimization Techniques
- Model Boosting Techniques
- Introduction to PyCaret
- Dealing with Unbalanced Datasets
- **4 Mini Projects**

Capstone Project- 3

Module 1 Module 2 **Module 3** Module 4 Module 5

Unsupervised Learning

- What is Unsupervised Learning?
- Application of Unsupervised Learning
- Introduction to Clustering
- Types of Clustering
- Partitioning Methods: K-means, DBSCAN, Spectral
- Hierarchical Methods: Hierarchical
- **2 Mini Projects**

Module 1 Module 2 **Module 3** Module 4 Module 5

Dimension Reduction

- PCA
- Factor Analysis
- LDA
- **Mini Project**

Module 1 Module 2 **Module 3** Module 4 Module 5

Association Rules Mining

- What are Association Rules?
- Association Rule Parameters
- A-priori Algorithm
- Market Basket Analysis

Module 1 Module 2 **Module 3** Module 4 Module 5

Recommendation System

- What is a Recommendation System?
- Need for a Recommendation System
- Recommendation System Use Cases
- Applications of Recommendation System
- Types of Recommendation Systems
- Collaborative Filtering
- Content Based Filtering
- Matrix Factorization
- Pros and Cons of Collaborative Filtering
- Hybrid Recommender System
- **Mini Project**

Module 1 Module 2 **Module 3** Module 4 Module 5

Time-series Forecasting

- Introduction to forecasting data
- Properties of Time Series data
- Features of Time Series data
- Markov Processes - Overview and Terminologies
- Naive, Average and Moving Average Forecasting
- Exponential Smoothing
- ARIMA Approach
- **Mini Project**

Module 1 Module 2 **Module 3** Module 4 Module 5

Machine Learning Model Deployment

- Overview of Machine Learning Models
- Machine Learning System Architecture
- Research Environment
- Packaging, Serving and Deploying the model
- Differential Testing And More...

Capstone Project- 4

Module 1 Module 2 Module 3 **Module 4** Module 5

Deep Learning & AI Specialization (57 Hours 19 Sessions)

Introduction to Deep Learning

- What is Deep Learning
- Curse of Dimensionality
- Machine Learning vs. Deep Learning
- Use Cases of Deep Learning
- Human Brain vs. Neural Network
- What is Perceptron?
- Learning Rate
- Epoch
- Batch Size

Module 1 Module 2 Module 3 **Module 4** Module 5

Computer Vision

- Introduction to Convolutional Neural Networks
- Introduction to images
- Convolution, Pooling, Padding & its mechanisms
- Forward Propagation & Backpropagation for CNNs
- CNN architectures
- AlexNet, VGGNet, InceptionNet & ResNet
- Transfer Learning
- Object Detection

Module 1 Module 2 Module 3 **Module 4** Module 5

Introduction to NLP

- Introduction to NLP
- Libraries & Tools for NLP in Python
- NLTK vs Spacy
- Applications of NLP:
- Chatbot
- Search, Autocorrect and Autocomplete
- Grammar Checker

Module 1 Module 2 Module 3 **Module 4** Module 5

Text Processing Methods

- Bag of Words
- Countvectorizer
- Term Frequency (TF)
- Inverse Document Frequency (IDF)
- Converting text to features and labels
- Multinomial Naive Bayes Classifier
- Leveraging Confusion Matrix Assignment
- Word Embeddings
- Word2Vec
- **Mini Project**

Module 1 Module 2 Module 3 **Module 4** Module 5

TensorFlow Hub for Object Detection using Faster RCNN

- Introduction to TensorFlow Hub
- Use cases of TensorFlow Hub
- Limitations of CNN in object detection
- Architecture of RCNN
- Applications of RCNN
- Types of RCNN
- Step by step implementation of Faster RCNN

Module 1 Module 2 Module 3 **Module 4** Module 5

Sentiment Analysis

- Sentiment Analysis
- Subjectivity Analysis
- Topic Extraction
- Product Reviews
- Opinion Retrieval and Spam
- Opinion Summarization
- Implementing Sentiment Analysis in Python

Module 1 Module 2 Module 3 **Module 4** Module 5

Reinforcement Learning

- Understanding Reinforcement Learning
- Algorithms associated with RL
- Q-Learning Model
- **Mini Project**

Module 1 Module 2 Module 3 **Module 4** Module 5

MLOps/LLMOps & Deployment

- Introduction To MLOps
- Automating Pipelines with Airflow
- Model Packaging
- API Building
- Model Training with AWS SageMaker
- Model Deployment
- Model Maintenance
- Git, Github and Github Actions
- CICD Pipeline
- Model packaging via Dockers and containers
- Deployment on DataBricks

Capstone Project- 6

Module 1 Module 2 Module 3 Module 4 **Module 5**

Advanced Generative AI, LLMs & Agentic AI Specialization (54 Hours 18 Sessions)

Introduction To Generative AI

- Introduction to Generative AI Models
- The Future of Generative AI
- Types of Generative AI Models
- Ethical Considerations in Generative AI Models & ChatGPT
- Popular Generative AI Models
- Benchmarking & Evaluating Models

Module 1 Module 2 Module 3 Module 4 **Module 5**

ChatGPT, Prompt Engineering, Explainable AI ChatGPT

Fundamentals of Prompt Engineering

- Zero-shot, Few-shot, Chain-of-Thought (CoT) Promptin
- Role-based Prompts, Multi-turn Conversations

Advanced Prompt Techniques

- System vs. User Prompts, Style Transfer, Iterative Prompting

Explainable AI

- Introduction to explainable AI
- Mini Project

Module 1 Module 2 Module 3 Module 4 **Module 5**

Advanced Generative AI Models & Architecture

- Introduction to Generative Models
- Large Language Models (LLMs)
- LLMs: GPT-4, DeepSeek, LLaMA, Mistral, Claude, Falcon
- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs) - StyleGAN, CycleGAN
- Attention Mechanisms and Transformers (GPT, BERT, T5)
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 **Module 5**

Advanced Generative AI - Building LLM Applications

- Langchain and Workflow Design
- Advanced Prompt Engineering Techniques
- LLM Application Development
- LLM Fine-Tuning and Customization
- Fine-Tuning & Pre-training of LLMs
- Benchmarking and Evaluation of LLM Capabilities
- LangChain for LLM Development
- RAG with Hugging Face & LangChain
- **2 Mini Projects**

Module 1 Module 2 Module 3 Module 4 **Module 5**

Agentic AI & Multimodal AI (Voice, Video & Document AI)

Agentic AI

- Frameworks
- Hugging Face Ecosystem
- Speech-to-Text & AI Voice Processing
- Whisper, Wav2Vec, DeepSeek-STT

Multimodal AI Text-to-Image & Image-to-Image Generation

- DALL·E, MidJourney, and Stable Diffusion
- LDMs & Hugging Face Implementations

Advanced Multimodal Models

- Google's Gemini, OpenAI's GPT-4o, and models from Meta AI
- Running ollama locally
- Deployemnt with FastAPI and Model Evaluation
- What MCP is and Why we need it?
- **Mini Project**

Module 1 Module 2 Module 3 Module 4 **Module 5**

Generative AI Applications In Industries

- Applications in Healthcare (e.g., Medical Imaging, Diagnosis)
- Applications in Finance (e.g., Risk Modeling, Fraud Detection)
- Applications in Retail and E-commerce (e.g., Personalized Marketing)
- Applications in Media and Entertainment (e.g., Content Creation)

Module 1 Module 2 Module 3 Module 4 **Module 5**

Responsible AI

- Awareness, Toxicity, Misinformation, Bias, Security, Fairness

Module 1 Module 2 Module 3 Module 4 **Module 5**

No-Code/Low-Code Agentic AI & AI Automation

- Introduction to AI Automation & No-Code AI
- n8n Fundamentals & Workflow Design
- AI-Powered Workflows using n8n
- LLM Integration with n8n
- AI Agents & Tool Calling in n8n

AI tools Covered

GitHub Copilot | CURSOR | Copilot

Gamma | Gemini | Lovable

Tools

jupyter | tableau | Midjourney | ChatGPT | DALL·E
Gemini | Google Cloud AutoML Vision | Amazon SageMaker | GitHub | kaggle
Power BI | perplexity | deepseek | git | OpenAI

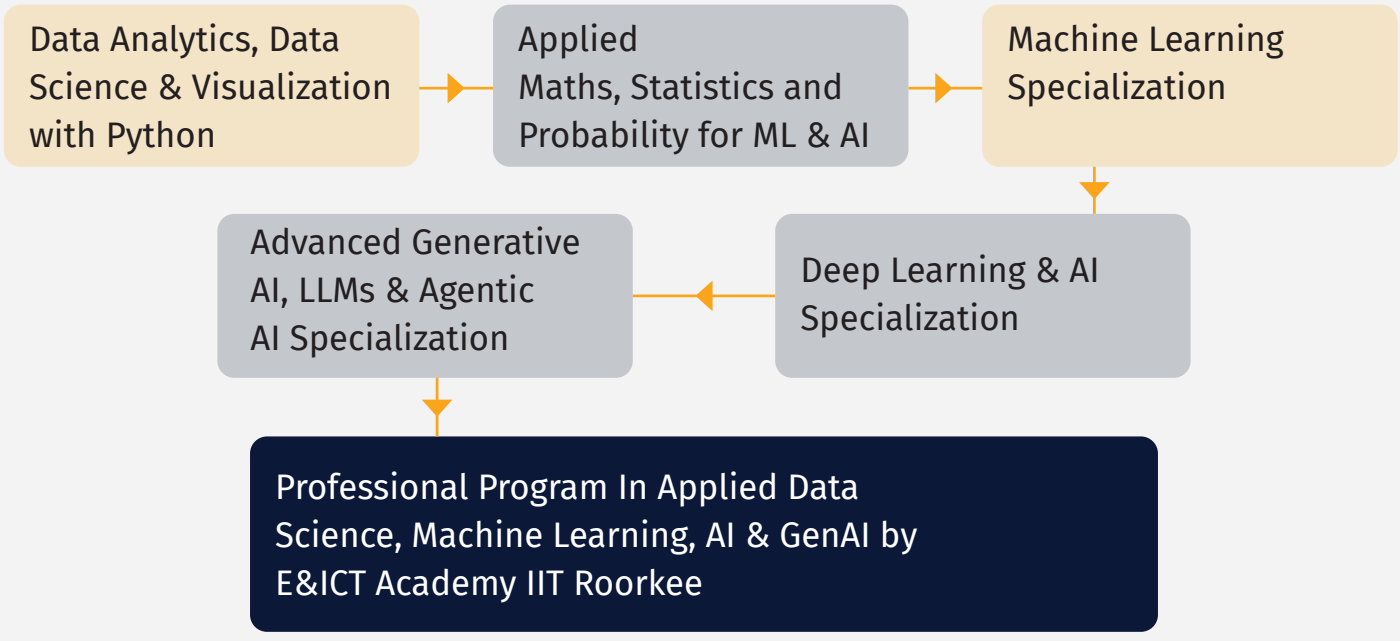
Languages

python | SQL | R

Frameworks

pandas | bokeh | matplotlib | OpenCV | spaCy
NumPy | scikit-learn | PyTorch | TensorFlow | MySQL
Keras | Hugging Face | TensorFlow Lite

Learning Path



E&ICT, IIT Roorkee

Professional Program In
Data Science, Machine Learning AI & GenAI

By E&ICT Academy, IIT Roorkee

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